

ORIGINAL

MODULE: SIC

MEMORY: $AC[18]$; $MD[18]$; $PC[13]$; $IR[13]$; $IA[13]$; $IB[13]$; $MA[13]$; lf ; $M[8193, 18]$

INPUTS: start

BUSES: $ABUS[18]$; $BBUS[18]$; $OBUS[19]$

1. $\rightarrow (\overline{start})/(1)$
2. $MA \leftarrow PC$
3. $MD \leftarrow \text{BUSFN}(M; \text{DCD}(MA))$
4. $IR \leftarrow MD$
5. $\rightarrow (IR_0 \wedge IR_1 \wedge IR_2)/(25)$
6. NO DELAY
 $\rightarrow ((\overline{IR_3} \wedge \overline{IR_4}), (\overline{IR_3} \wedge IR_4), IR_3)/(13, 7, 10)$
7. $MA \leftarrow IR_{5:17}$
8. $MD \leftarrow \text{BUSFN}(M; \text{DCD}(MA))$
9. $IR_{5:17} \leftarrow MD_{5:17}$
 $\rightarrow (13)$
10. NO DELAY
 $\rightarrow (IR_4)/(12)$
11. $IR_{5:17} \leftarrow \text{ADD}(IR_{5:17}; IA)$
 $\rightarrow (13)$
12. $IR_{5:17} \leftarrow \text{ADD}(IR_{5:17}; IB)$
13. NO DELAY
 $\rightarrow (\overline{IR_0} \wedge \overline{IR_1})/(15)$
14. $PC \leftarrow IR_{5:17}$
 $\rightarrow (2)$
15. $MA \leftarrow IR_{5:17}$
 $\rightarrow (IR_0)/(21)$
16. $MD \leftarrow \text{BUSFN}(M; \text{DCD}(MA))$
 $\rightarrow (\overline{IR_1} \wedge \overline{IR_2})/(18)$
17. $AC \leftarrow (MD! (MD \wedge AC)! (\text{ADD}_{1:18}(MD; AC))) * ((\overline{IR_1} \wedge IR_2), (IR_1 \wedge \overline{IR_2}), (IR_1 \wedge IR_2))$
 $lf * (IR_1 \wedge IR_2) \leftarrow \text{ADD}_0(MD; AC)$
 $\rightarrow (24)$
18. $MD \leftarrow \text{INC}(MD)$
19. $M * \text{DCD}(MA) \leftarrow MD$
 $\rightarrow (\vee / MD)/(24)$
20. $PC \leftarrow \text{INC}(PC)$
 $\rightarrow (24)$
21. $MD \leftarrow ((AC)! (5 \uparrow 0, \text{INC}(PC))) * (IR_2, \overline{IR_2})$
22. $M * \text{DCD}(MA) \leftarrow MD$
 $\rightarrow (IR_2)/(24)$
23. $PC \leftarrow IR_{5:17}$
24. $PC \leftarrow \text{INC}(PC)$
 $\rightarrow (2)$
25. NO DELAY
 $\rightarrow (IR_3)/(50)$
26. NO DELAY
 $\rightarrow (IR_5)/(30)$

27. NO DELAY
 $\rightarrow ((IR_6 \wedge IR_7), (\overline{IR_6 \wedge IR_7}))/ (1, 29)$
 (No Step 28)
29. $AC * (IR_8 \vee IR_9) \leftarrow ((\overline{18 \uparrow 0}))! (18 \uparrow 0)! \overline{AC}$
 $* ((\overline{IR_8 \wedge IR_9}), (IR_8 \wedge \overline{IR_9}), (IR_8 \wedge IR_9));$
 $If * IR_6 \leftarrow 0; If * IR_7 \leftarrow 1$
 $\rightarrow (33)$
30. $\rightarrow (IR_4)/ (32)$ NO DELAY
31. $If, AC \leftarrow AC, If$
 $\rightarrow (33)$
32. $If, AC \leftarrow AC_{17}, If, AC_{0:16}$
33. NO DELAY
 $\rightarrow (IR_{10})/ (40)$
34. NO DELAY
 $\rightarrow ((\overline{IR_{11} \wedge \overline{IR_{12}}}), (\overline{IR_{11} \wedge IR_{12}}), (IR_{11} \wedge \overline{IR_{12}}),$
 $(IR_{11} \wedge IR_{12}))/ (35, 37, 38, 39)$
35. NO DELAY
 $\rightarrow (\overline{IR_{13} \wedge \overline{If}})/ (43)$
36. $PC \leftarrow INC(PC)$
 $\rightarrow (24)$
37. $AC \leftarrow (5 \uparrow 0, IA! 5 \uparrow 0, IB) * (\overline{IR_{13}}, IR_{13})$
 $\rightarrow (43)$
38. $IA \leftarrow ((AC_{5:17})! (INC(IA))) * (\overline{IR_{13}}, IR_{13})$
 $\rightarrow (43)$
39. $IB \leftarrow ((AC_{5:17})! (INC(IB))) * (\overline{IR_{13}}, IR_{13})$
 $\rightarrow (43)$
40. NO DELAY
 $\rightarrow (IR_4)/ (42)$
41. $If, AC \leftarrow AC, If$
 $\rightarrow (43)$
42. $If, AC \leftarrow AC_{17}, If, AC_{0:16}$
43. NO DELAY
 $\rightarrow (IR_{14})/ (45)$
44. $\rightarrow (f, \overline{f})/ (36, 24)$
45. NO DELAY
 $\rightarrow (IR_4)/ (47)$
46. $If, AC \leftarrow AC, If$
 $\rightarrow (24)$
47. $If, AC \leftarrow AC_{17}, If, AC_{0:16}$
 $\rightarrow (24)$

END SEQUENCE

$$f = (AC_0 \wedge IR_{15}) \vee ((\vee / AC) \wedge IR_{16}) \vee (\overline{AC_0} \wedge (\vee / AC) \wedge IR_{17})$$

END.

MODULE: SIC

MEMORY: $AC[18]; MD[18]; PC[13]; IR[13]; IA[13]; IB[13]; MA[13]; If; M[8193, 18]$

INPUTS: start

BUSES: ABUS [18]; BBUS [18]; OBUS [19]

INICIO {

1. $\rightarrow (\overline{start})/(1)$
2. $MA \leftarrow PC$
3. $MD \leftarrow \text{BUSFN}(M; \text{DCD}(MA))$
4. $IR \leftarrow MD$
5. $\rightarrow (IR_0 \wedge IR_1 \wedge IR_2)/(25) \leftarrow \text{Excluye Ins. Operativas (111)}$
6. NO DELAY
 $\rightarrow ((\underbrace{\overline{IR_3} \wedge \overline{IR_4}}_{\text{DIR.}}, \underbrace{\overline{IR_3} \wedge IR_4}_{\text{INDIR.}}, \underbrace{IR_3}_{\text{INDEX.}})/(13,7,10)$

INDIRECTO {

- 7. $MA \leftarrow IR_{5:17}$ ← Ponemos dirección indirecta
- 8. $MD \leftarrow \text{BUSFN}(M; \text{DCD}(MA))$ ← Sacamos contenido
- 9. $IR_{5:17} \leftarrow MD_{5:17}$ ← Obtenemos dirección real
→ (13)

IRO	IR1	IR2	NEMONICO
0	0	0	ISZ
0	0	1	LAC
0	1	0	AND
0	1	1	TAD
1	0	0	JMS
1	0	1	DAC
1	1	0	JMP
1	1	1	OPERATIVAS

			LAC	AND	TAD
OPERACIÓN "ALU"	{	17.	$AC \leftarrow \left(MD! (MD \wedge AC)! (ADD_{1:18}(MD; AC)) \right) * ((\overline{IR_1} \wedge IR_2), (IR_1 \wedge \overline{IR_2}), (IR_1 \wedge IR_2))$ $lf * (IR_1 \wedge IR_2) \leftarrow ADD_0(MD; AC) \leftarrow \text{Actualizar LF con TAD}$ $\rightarrow (24)$		
ISZ		18.	$MD \leftarrow INC(MD) \leftarrow \text{Incrementar el contenido}$		
	{	19.	$M * DCD(MA) \leftarrow MD \leftarrow \text{Guarda el nuevo contenido}$		
		20.	$\rightarrow (V/M D)/(24) \leftarrow \text{Verifica si es igual a 0 y si no va al paso 24 para finalizar}$ $PC \leftarrow INC(PC) \leftarrow \text{Si es 0 incrementa el puntero y luego va al paso 24 para terminar}$ $\rightarrow (24)$		
ESCRITURA	{	21.	DAC	JMS	$MD \leftarrow ((AC)! (5 \uparrow 0, INC(PC))) * (IR_2, \overline{IR_2})$
		22.	$M * DCD(MA) \leftarrow MD \leftarrow \text{Escribimos en la memoria}$ $\rightarrow (IR_2)/(24) \leftarrow \text{Bifurcación para el nuevo puntero de JMS}$		
		23.	$PC \leftarrow IR_{5:17} \leftarrow \text{Carga de la posición de salto del puntero}$		
		24.	$PC \leftarrow INC(PC) \leftarrow \text{Bifurcación para el nuevo puntero de JMS}$ $\rightarrow (2)$		

EXCLUSION { 25. NO DELAY
I/O → $(IR_3)/(50)$

PRIMER EVENTO

DECISION { 26. NO DELAY
ROTACION → $(IR_5)/(30)$ ← Rotar IR5=1 o NO rotar IR5=0
CASO NOP { 27. NO DELAY
→ $((\overline{IR_6 \wedge IR_7}), (\overline{IR_6 \wedge IR_7}))/(\underline{1}, 29)$
HLT

(No Step 28)

AJUSTE { 29. $AC * (IR_8 \vee IR_9) \leftarrow ((18 \uparrow 0))! (18 \uparrow 0)! \overline{AC}$
ACUMULADOR * $((\overline{IR_8 \wedge IR_9}), (IR_8 \wedge \overline{IR_9}), (IR_8 \wedge IR_9));$
CLL STL
 $If * IR_6 \leftarrow 0; If * IR_7 \leftarrow 1$
→ (33)
ROTACIONES { 30. → $(IR_4)/(32)$ NO DELAY
31. $If, AC \leftarrow AC, If$
→ (33)
32. $If, AC \leftarrow AC_{17}, If, AC_{0:16}$

SEGUNDO EVENTO

DECISION { 33. NO DELAY
ROTACION → $(IR_{10})/(40)$ ← Rotar IR10=1 o NO rotar IR10=0
34. NO DELAY

NOP/SZL DFA/DFB DTA/INA DTB/INB
→ $((\overline{IR_{11} \wedge IR_{12}}), (\overline{IR_{11} \wedge IR_{12}}), (IR_{11} \wedge \overline{IR_{12}}), (IR_{11} \wedge IR_{12}))/ (35, 37, 38, 39)$

NOP/SZL { 35. NO DELAY
→ $(IR_{13} \wedge \overline{If})/(43)$ ← Verifica si LF=0 sino va al siguiente evento
36. $PC \leftarrow INC(PC)$ ← Si es 0 incrementa el puntero y luego va al paso 24 para terminar
→ (24) DFA/DFB

DFA/DFB { 37. $AC \leftarrow (5 \uparrow 0, IA \uparrow 5 \uparrow 0, IB) * (\overline{IR_{13}}, IR_{13})$
→ (43)

DTA/INA { 38. $IA \leftarrow ((AC_{5:17})! (INC(IA))) * (\overline{IR_{13}}, IR_{13})$
→ (43)

DTB/INB { 39. $IB \leftarrow ((AC_{5:17})! (INC(IB))) * (\overline{IR_{13}}, IR_{13})$
→ (43)

ROTACIONES { 40. NO DELAY
→ $(IR_4)/(42)$
41. $If, AC \leftarrow AC, If$
→ (43)
42. $If, AC \leftarrow AC_{17}, If, AC_{0:16}$

TERCER EVENTO

DECISION { 43. NO DELAY
ROTACION → $(IR_{14})/(45)$
SALTOS { 44. → $(f, \overline{f})/(36, 24)$

ROTACIONES $\left\{ \begin{array}{l} 45. \text{ NO DELAY} \\ \rightarrow (IR_4)/(47) \\ 46. If, AC \leftarrow AC, If \\ \rightarrow (24) \\ 47. If, AC \leftarrow AC_{17}, If, AC_{0:16} \\ \rightarrow (24) \end{array} \right.$

END SEQUENCE

$$f = \underbrace{(AC_0 \wedge IR_{15})}_{AC < 0} \vee \underbrace{(\overline{(V/AC)} \wedge IR_{16})}_{AC = 0} \vee \underbrace{(\overline{AC_0} \wedge (V/AC) \wedge IR_{17})}_{AC > 0}$$

END.